

5 Normal Multivariate Distribution

MA2003B Application of Multivariate Methods
in Data Science

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Definition and Basic Properties

Definition

Let μ be a (row) vector and let Σ be a symmetric positive definite matrix. We say that a random (row) vector X has **normal multivariate distribution** with mean μ and covariance matrix Σ , in symbols $X \sim \mathcal{N}_p(\mu, \Sigma)$, if its probability density is given by

$$\frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(X - \mu)\Sigma^{-1}(X - \mu)^T\right) dX$$

Remark

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4. Notice that the term $(X - \mu)\Sigma^{-1}(X - \mu)^T$ is the square of the Mahalanobis distance between the random vector X and the mean μ .

Just like in the univariate case, the normal multivariate distribution is singled out among all other possible distributions by the Central Limit Theorem, that roughly states that the sum of a large number of independent and identically distributed random variables (properly standardized) converges to a normal distribution.

Multivariate Central Limit Theorem

Theorem

Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of independent and identically distributed random vectors in $\mathbb{R}^p$ with mean vector μ and covariance matrix Σ . Define the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then, as $n \rightarrow \infty$,

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}_p(0, \Sigma),$$

where $\xrightarrow{d}$ denotes convergence in distribution and $\mathcal{N}_p(0, \Sigma)$ is the p -dimensional multivariate normal distribution with mean vector 0 and covariance matrix Σ .

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Furthermore, important statistical methods, like linear regression, principal component analysis and clustering do not yield good results when applied to a normal multivariate distribution.

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The probability distribution of these contours is controlled by the χ^2 distribution in the following sense:

Let $0 \leq \varpi \leq 1$ be a probability level, and let c be such that

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Then, the contour that contains a proportion ϖ of the probability mass of the distribution is given by the set of points X that satisfy the equation

$$(X - \mu)\Sigma^{-1}(X - \mu)^T = c$$

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The color gradient represents the density of the distribution, with darker colors indicating higher density.

Contours of the Bivariate Normal

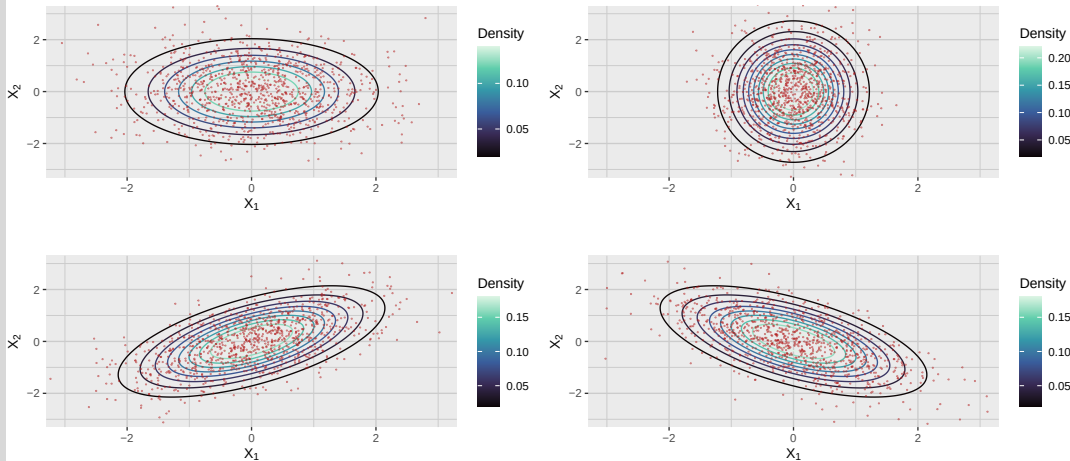


Figure 1: Bivariate Normal Distribution with Different Covariance Matrices

Standardization of the Normal Multivariate Distribution

Just like in the univariate case, given a random vector $X \sim \mathcal{N}_p(\mu, \Sigma)$, we can standardize it to obtain a new random vector Z that follows the standard multivariate normal distribution $\mathcal{N}_p(0, I_p)$, where I_p is the identity matrix of size p .

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To describe this standardization process, we first need to remember some linear algebra results.

Singular Value Decomposition

Theorem

Let Σ be a symmetric positive definite matrix of size $p \times p$. Then, there exists an orthogonal matrix Q and a diagonal matrix Λ with positive entries such that

$$\Sigma = Q\Lambda Q^T$$

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Remark

Remember that a matrix Q is orthogonal if $Q^T Q = I_p$, where I_p is the identity matrix of size p . This means that the columns of Q are orthonormal vectors, and thus they form a basis for $\mathbb{R}^p$.

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In the case of the covariance matrix Σ , the SVD provides a way to express it in terms of its **singular values** (the diagonal entries of Λ , also known as its **eigenvalues**) and its **singular vectors** (the columns of Q , also known as its **eigenvectors**).

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This decomposition is particularly useful for understanding the geometry of the multivariate normal distribution.

Notice that, since Σ is positive definite, the diagonal entries of

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_p \end{bmatrix}$$

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are all positive.

In particular, we can define a new matrix

$$D = \begin{bmatrix} \sqrt{\lambda_1} & 0 & \cdots & 0 \\ 0 & \sqrt{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{\lambda_p} \end{bmatrix}$$

satisfying $D^2 = \Lambda$.

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In particular, we can define the **standardization** of the random vector X as

$$Z = (X - \mu)\sigma^{-1} = (X - \mu)QD^{-1} = (X - \mu)Q\Lambda^{-1/2}$$

in which case $Z \sim \mathcal{N}_p(0, I_p)$.

Closure under Linear Transformations

Theorem

Let $X \sim \mathcal{N}_p(\mu, \Sigma)$ be a random vector with normal multivariate distribution. Then, for any matrix A of size $m \times p$ and any vector b of size m , the random vector

$$Y = AX + b$$

also has a normal multivariate distribution, specifically $Y \sim \mathcal{N}_m(A\mu + b, A\Sigma A^T)$.

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In practice, this theorem is often used to derive the distribution of linear combinations of random variables, which is a common operation in multivariate statistics.

Example

Let $X \sim \mathcal{N}_2(\mu, \Sigma)$ be a random vector with mean $\mu = (1, 2)$ and covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 2 \end{bmatrix}$$

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In other words,

$$Y_1 = 2X_1 + X_2$$

$$Y_2 = 3X_2 + 1$$

Example

Using the closure under linear transformations theorem, we can find the mean and covariance of Y :

$$\text{Mean: } A\mu + b = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

$$\text{Covariance: } A\Sigma A^T = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0.5 \\ 0.5 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 9 \\ 9 & 18 \end{bmatrix}$$

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Thus,

$$Y \sim \mathcal{N}_2 \left(\begin{bmatrix} 4 \\ 7 \end{bmatrix}, \begin{bmatrix} 8 & 9 \\ 9 & 18 \end{bmatrix} \right)$$

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This is a very useful property, as it allows us to analyze subsets of variables while still maintaining the normality assumption.

Multivariate Normality Testing

There are several statistical tests to check whether a given dataset follows a multivariate normal distribution. These tests are essential in multivariate statistics, as many methods assume normality. Here are some of the most commonly used tests, together with their implementation in R:

1. **Mardia's Test:** Based on multivariate skewness and kurtosis. For a multivariate normal distribution, skewness should be close to 0 and kurtosis close to $p(p+2)$, where p is the number of variables. Implemented in R via `MVN::mvn(your_data, mvn_test = "mardia")`.
2. **Henze-Zirkler (HZ) Test:** Uses the empirical characteristic function to measure deviations from normality. It has good global power and is suitable for moderate dimensions. Implemented in R via `MVN::mvn(your_data, mvn_test = "hz")`.

3. **Royston's Test:** A multivariate extension of the Shapiro-Wilk test, based on the product of univariate Shapiro-Wilk statistics applied to principal components. Easy to interpret, though power decreases in high dimensions. Implemented in R via `MVN::mvn(your_data, mvn_test = "royston")`.
4. **Doornik-Hansen Test:** Combines skewness and kurtosis into a single omnibus test, providing a robust alternative to Mardia. Implemented in R via `MVN::mvn(your_data, mvn_test = "dh")`.
5. **Graphical Methods (Chi-square Q-Q plots):** Plot the Mahalanobis distances of the data against the quantiles of a chi-square distribution. Deviations from a straight line indicate departures from multivariate normality. Implemented in R via `multivariate_diagnostic_plot(data = your_data, type = "qq")` .